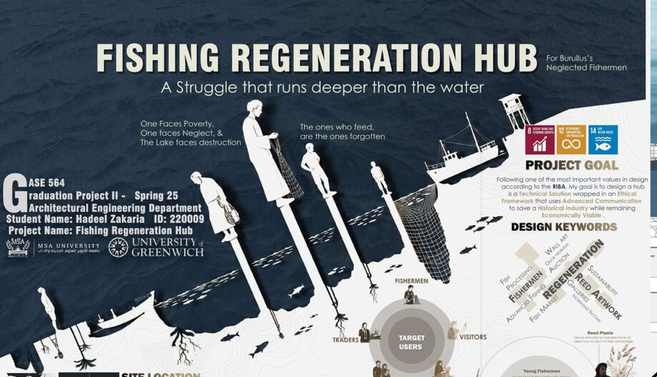
FISHING REGENERATION HUB

For Burullus's Neglected Fishermen

A Struggle that runs deeper than the water

One Faces Poverty,
One faces Neglect, &
The Lake faces destruction

The ones who feed,
are the ones forgotten



GASE 564
Graduation Project II – Spring 25
Architectural Engineering Department
Student Name: Hadeel Zakaria ID: 220009
Project Name: Fishing Regeneration Hub

MSA University | University of Greenwich

PROJECT GOAL

Following one of the most important values in design according to the **RIBA**, My goal is to design a hub is a Technical Solution wrapped in an Ethical framework that uses Advanced Communication to save a Historical Industry while remaining Economically Viable

DESIGN KEYWORDS

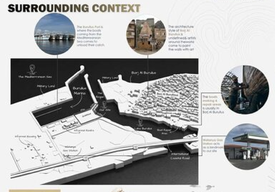
SITE LOCATION

The Bogaz in Lake Burullus, Kafr el Sheikh (often referred to as Bughaz El-Burullus) is the vital, narrow natural channel that connects the Mediterranean Sea to Lake Burullus, the second-largest natural lake in Egypt's Nile Delta. It serves as the primary, direct connection between the two water bodies, acting as the main artery for the exchange of water, fish, and nutrients. The site has a relatively narrow sandbar structure (which separates the lake from the sea) and acts as the gateway for seawater into the predominantly brackish, shallow lake. The sea is deeper than the surrounding lake, with a maximum depth reaching 4 meters.

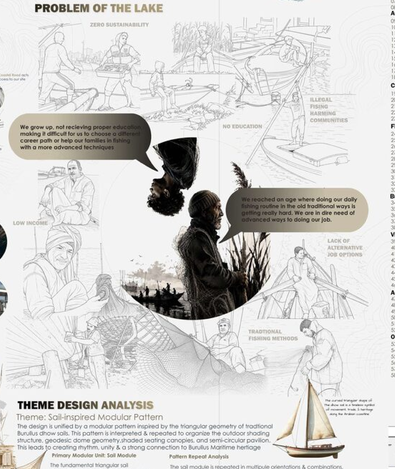
VIEWS INTO SITE

ACCESSIBILITY

SURROUNDING CONTEXT



PROBLEM OF THE LAKE



CONCEPT: Making the Invisible Visible.

Burullus does not stand empty; it breathes quietly through salt-stained wood, through the rhythm of casting nets at dawn, through reeds bending with the wind along the bogaz. Its identity has never been absent, only unspoken, dissolved into the ordinary. Our Fishing Regeneration Hub emerges from this silence as an act of revelation. It gathers the unnoticed textures of fishermen's lives, the grain of boat timber, the geometry of woven nets, the crystallized trace of salt, and transforms them into space, structure, and light. Here, architecture does not invent meaning; it uncovers it. It makes the invisible visible, framing labor as heritage, routine as ritual, and community as culture. In doing so, Burullus is no longer a place passed by. It becomes a place recognized, by its people and by the world.

THEME DESIGN ANALYSIS

Theme: Sail-inspired Modular Pattern

The design is unified by a modular pattern inspired by the triangular geometry of traditional Burullus straw sails. This pattern is interpreted & repeated to organize the outdoor shading structure, geodesic dome geometry,shaded seating canopies, and semi-circular pavilion. This leads to creating rhythm, unity & a strong connection to Burullus Maritime heritage

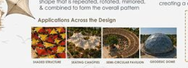
Primary Modular Unit: Sail Module

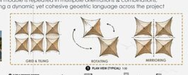
The fundamental triangular sail shape that is repeated, rotated, mirrored, & combined to form the overall pattern

Pattern Repeat Analysis

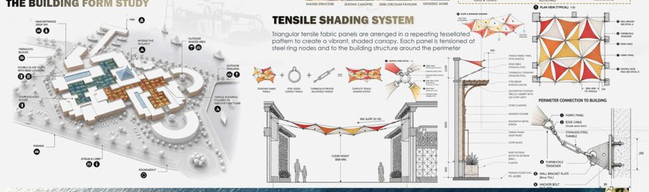
The sail module is repeated in multiple orientations & combinations, creating a dynamic yet cohesive geometric language across the project

Applications Across the Design



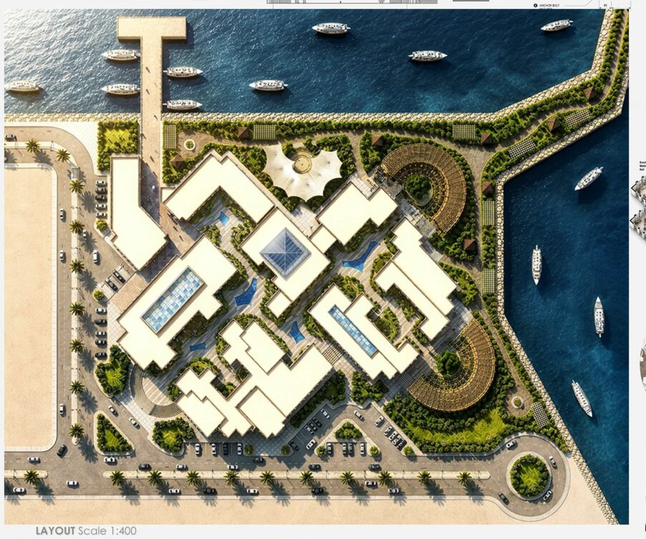


THE BUILDING FORM STUDY



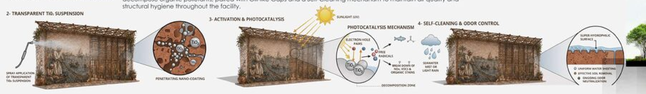
TENSILE SHADE SYSTEM

Triangular tensile fabric panels are arranged in a repeating tessellated pattern to create a vibrant shaded canopy. Each panel is tensioned at steel ring nodes and to the building structure around the perimeter



ODOR TREATMENTS

The building skin employs a transparent TiO_2 Suspension that undergoes Photocatalysis under sunlight to decompose organic pollutants, paired with Gill-like Gaps and a Self-Cleaning mechanism to maintain air quality and structural hygiene throughout the facility.



LEGEND

Fishermen's Zone
01. Lockers Rooms
02. Restrooms
03. Kitchenette
04. Changing Room
05. Fishermen's Entrance
06. Fishing Equipment Storage
07. Fish Processing Zone
08. Boat Fixing Equipments
Auction Zone
09. Trader's Entrance
10. Auction Hall
11. Loading Zone
12. Fish Order Preparation
13. Cold Refrigerators
14. Fish Packaging Storage
15. Cashier Area
16. Quality Inspection Office
17. Managerial Offices
18. Traders waiting Hall
Community Training Zone
19. Managerial Training Offi
20. First Aid Training Clinic
21. Seminar Halls
22. Reed Making Training Ha
23. Artwork Training Rooms
Fish Market Zone
24. Harbour Security
25. Clinic
26. Food Take-away Prep
27. Kitchen
28. Dehydrator Zone
29. Kitchen Storage
30. Fish Market
31. Restrooms Waiting Hall
32. Meeting Room
33. Restrooms
Burullus Cultural Zone
34. Reed Cafe
35. Reed Market
36. Artwork Gift Shops
37. Art Workshops
38. Restrooms
Visitors Zone
39. Reception
40. Lobby
41. Guest Waiting Room
42. Temporary Gallery
43. Permenant Gallery
44. Theatre Hall
45. Storage
46. Preservist Entrance
Administrative Departme
47. Ticket Office
48. Admins Office
49. Meeting Rooms
50. Storage
51. Electrical Room
52. Hub Maintenance Office
Outdoor Zone
53. Open Pavilion
54. Event Tensile area
55. Promenade
56. Community Parking
57. Loading Trucks Parking
58. Jetty (Pier)

MASTERPLAN Scale 1:200

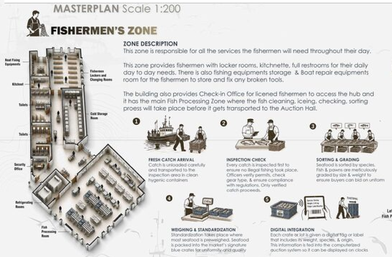
FISHERMEN'S ZONE

ZONE DESCRIPTION

This zone is responsible for all the services the fishermen will need throughout their day.

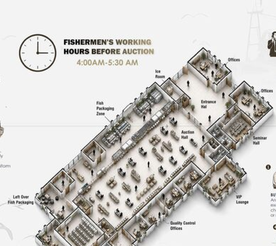
This zone provides fishermen with locker rooms, kitchenette, full restrooms for their daily day to day needs. There is also fishing equipments storage & boat repair equipments room for the fishermen to store and fix any broken tools.

The building also provides Check-In Office for licened fishermen to access the hub and it has the main Fish Processing Zone where the fish cleaning, iceing, checking, sorting prcess will take place before it gets transported to the Auction Hall.



FISHERMEN'S WORKING HOURS BEFORE AUCTION

4:00AM-5:30 AM



AUCTION ZONE

ZONE DESCRIPTION

This zone is responsible for Auction Hall and all the managerial services around it.

This zone welcomes the traders at 5:30 AM to let them check the fresh fish in person, as they will then register at the Auctioneer office to enter the bidding process.

The rest of the building include waiting lounge for the traders to wait for the bidding to start, managerial offices and on-site management training for young boys, quality control offices, electrical offices, cashier, and fish packaging and icing zone.

AUCTION TIME

5:30 AM - 9:00 AM

FISH MARKET ZONE

ZONE DESCRIPTION

This zone is responsible for Fish Market & Restaurant.

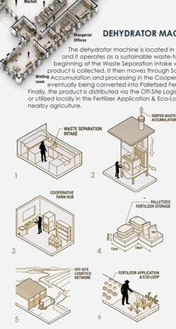
This zone acts as the main meeting point between all users as everyone can access it anytime.

The building has a main fish market that has an interactive skylight that helps in reducing heat and odor to the fish market The back kitchen and food-takeway preparation zone are behind the market.

INTERACTIVE SKYLIGHT

DEHYDRATOR MACHINE

The dehydrator machine is located in the food prep zone and it operates as a sustainable waste-to-resource loop, beginning at the Waste Separation Intake where organic by-product is collected. It then moves through Sorted Waste Accumulation and processing in the Cooperative Farm Hub, eventually being converted into Palletized Fertilizer Storage. Finally, the product is distributed via the Off-Site Logistics Network for agriculture.



BURULLUS CULTURAL ZONE

ZONE DESCRIPTION

This zone is responsible for all Reed-making artwork and normal household products.

This zone trains the community on how to utilize the reed plants to their advantage wo/thout harming the environment. This training not also opens doors for job opportunities apart from fishing but also increase the family's income, thus improve life in a sustainable way.

The building also has a Reed-cafe where locals and visitors can have coffee and bakries whilst making reeds with each lother.

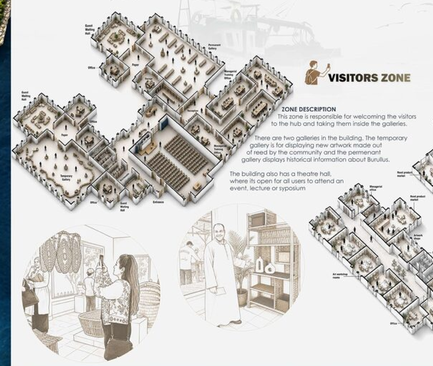
VISITORS ZONE

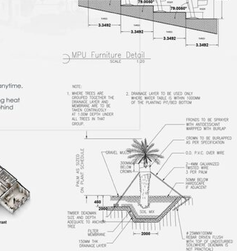
ZONE DESCRIPTION

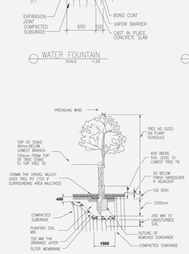
This zone is responsible for welcoming the visitors to the hub and taking them inside the galleries.

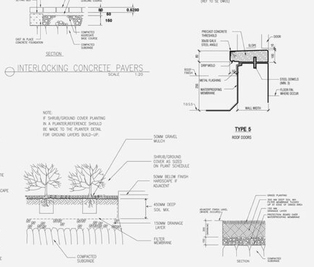
There are two galleries in the building. The temporary gallery is for displaying new artwork made out of reed by the community and the permenant gallery displays historical information about Burullus.

The building also has a theathe hall, where its open for all users to attend an event, lecture or symposium



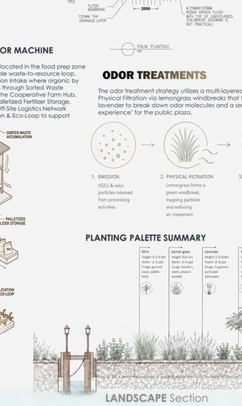






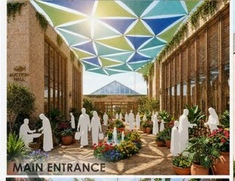
STRUCTURAL DETAILS

HOLLOW BLOCKS
PANELLED BEAMS
FRAMES



ODOR TREATMENTS

The odor treatment strategy utilizes a multi-layered filtration system to ensure a 'fresh environment for all users, starting with Physical Filtration via lemongrass windbreaks that trap airborne particles. This is followed by Chemical Neutralization using lavender to break down odor molecules and a sensory Reset through mint, which masks lingering scents to provide a 'fresher experience' for the public plaza.

PLANTING PALETTE SUMMARY

HOW IT WORKS: THE OLFACTORY FILTERATON PROCESS